

Case Study: Job Change Prediction

Context:

- An Ed-Tech company wants to hire data scientists among people who have successfully passed some courses and then signed up for training. The company wants to know which of the people are really looking for a job change and will prefer working with them, after completion of training because it helps to reduce the cost and time for categorization of candidates.

Information related to demographics, education, experience is in hands from candidates signup and enrollment.

Problem:

The data-set aims to answer the following key questions:

- What are the factors affecting a person looking for a job change?
- Does imbalance in the data affect model predictions?
- What are the key business recommendations based on analysis and model?

Attribute Information:

- The data contains characteristics of the people
 - **enrollee_id**: Unique ID for the candidate
 - **city**: City code
 - **city_development_index**: Development index of the city (scaled)
 - **gender**: Gender of the candidate
 - **relevent_experience**: Relevant experience of the candidate in years

- **major_discipline:** Education major discipline of the candidate
- **experience:** Candidate total experience in years
- **company_size:** No of employees in current employer's company
- **company_type:** Type of current employer
- **lastnewjob:** Difference in years between previous job and current job
- **training_hours:** training hours completed
- **target:** 0 – Not looking for a job change, 1 – Looking for a job change

Learning Outcomes:

- Exploratory Data Analysis
- K-fold Cross-validation
- Handling imbalanced data
- Regularization to reduce overfitting

Steps and Tasks:

- Import Libraries and Load Dataset
- Overview of data
- Data Visualization
- Data preparation
- Choose model
- Over and under-sample train set to balance the classes
- Perform regularization, if needed
- Conclusion

